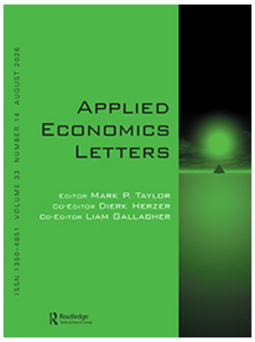

Applied Economics Letters

ISSN: 1350-4851 (Print) 1466-4291 (Online) Journal homepage: www.tandfonline.com/journals/rael20

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APPLIED ECONOMICS LETTERS
2026, VOL. 33, NO. 14, 2441–2445
https://doi.org/10.1080/13504851.2025.2473662

Routledge
Taylor & Francis Group

How city size affects firm survival: evidence from Chinese enterprise registration data

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ABSTRACT
Using firm-level registration data and city-level statistics, this paper empirically investigates the causal relationship between city size and firm survival. The results reveal that firms in larger cities exhibit higher survival rates. Mechanism analysis suggests that larger cities create positive externalities – such as increased openness level, lower entry cut-off, and enhanced innovation capacity – that help mitigate the adverse effects of higher wages and costs. However, this positive relationship does not increase monotonically. We identify an inverted U-shaped relationship between Chinese city size and firm survival from 2001 to 2014, with an optimal city size of 5.59 million people, beyond which congestion costs begin to outweigh the agglomeration benefits.

KEYWORDS
City size; firm survival; agglomeration benefit; congestion cost

JEL CLASSIFICATION
R12; L25; O18

I. Introduction

Cities, as hubs of economic activity, shape firm survival through agglomeration economies and competitive pressures. While larger cities offer firms access to expansive consumer markets, diversified labour pools, dense knowledge spillover networks, and robust upstream-downstream industrial linkages – factors that enhance survival prospects by lowering operational risks and fostering innovation[1] —excessive agglomeration may amplify congestion costs through rising wages, land prices, and homogenized competition.[2] This paradox raises a critical question: Does an optimal city size exist that balances agglomeration benefits and congestion costs to maximize firm survival?

Our investigation builds upon two established research streams. The first examines city size distribution and its economic implications, tracing from early formulations of Pareto distributions (Singer 1936) and Zipf's Law (Zipf 1949) to modern analyses of optimal city size. Seminal work by Henderson (1974) established the inverted U-shaped relationship between city size and productivity, a framework extended by Duranton and Puga (2020) who synthesized agglomeration's dual effects on firm dynamics. Recent contributions by Duranton and Puga (2023) and Turner and Weil (2025) further contextualize cities' roles as innovation catalysts and growth engines. The second strand focuses on firm survival determinants, distinguishing between external drivers such as FDI spillovers (Helmers and Rogers 2010) and crisis shocks (Iwasaki 2014), and internal factors including firm age and technological capability (Agarwal and Gort 2002; Boyer and Blazy 2014). It also demonstrates how urbanization economies enhance survival through productivity gains (Combes et al. 2012), although localization effects may counteract such benefits (Burger, Van Oort, and Raspe 2011). Despite these advances, research on how city size impacts firm survival remains limited. Prior studies predominantly focus on macro-level productivity outcomes, leaving the micro-level dynamics of firm survival underexplored. The literature also lacks consensus on how congestion effects interact with agglomeration forces at different urban scales.

Our study makes three key contributions to address these limitations. First, by utilizing firm-level registration data alongside city-level statistics, we provide a systematic examination of the

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[1]See Florida (2002); Rosenthal and Strange (2003); Glaeser (2008); Combes et al. (2012); Roca and Puga (2017); Grover, Lall, and Timmis (2023).

[2]See Henderson (1974); Ciccone and Hall (1996); Duranton and Puga (2004); Carlino, Chatterjee, and Hunt (2007); Combes et al. (2010).

Supplemental data for this article can be accessed online at https://doi.org/10.1080/13504851.2025.2473662

Table 1. Baseline results.

	(1)	(2)
Dependent variable	Firm survival rate	
City size	0.007***	0.033*
	(0.002)	(0.019)
Observations	3471	3150
Controls	No	Yes
City FE	No	Yes
Time FE	No	Yes
Adj. R^2	0.004	0.107

Note: The brackets denote the standard error clustered at the city level; $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

nonlinear relationship between urban size and firm survival – quantifying an inverted U-shaped curve with an optimal city population threshold of 5.59 million. Second, we elucidate the mechanisms driving this relationship – such as openness, innovation incentives, and productivity cut-offs. Third, we resolve the conflicting evidence on agglomeration effects by showing how the survival benefits of urban scale shift from positive to negative once congestion costs start to dominate.

II. Strategy

Estimation methodology

To identify the impact of city size on firm survival, this paper sets up the model as follows:

$$Firm\ survival\ rate_{it} = \alpha + \beta City\ size_{it} + \delta X_{it} + \mu_i + \lambda_t + \varepsilon_{it}, \tag{1}$$

where $Firm\ survival\ rate_{it}$ reflects the firm survival rate in sample city i during period t; $City\ size_{it}$ is the logarithm of population of city i in year t; X_{it} represents all other control variables for city i at different times. μ_i represents city-fixed effects, λ_t is the time-fixed effect; ε_{it} is the random disturbance term.[3]

III. Empirical evidence

Baseline results

According to column (1) in Table 1, we find that the relationship between city size and the survival rate of firms is positive and statistically significant. In column (2), after adding control variables and two-way fixed effects, the results remain significantly positive. It shows that for every 1% increase in city size, the survival rate increases by 0.03%. That is to say, city size plays a positive role in promoting firm survival, larger cities are more conducive to the survival of firms, supporting the theory of agglomeration benefits.

We address potential endogeneity concerns through multiple strategies, including the use of one-period lagged variables, controlling for failure contagion effects, employing night-time light data as a proxy for the dependent variable, leveraging several instrumental variables, and excluding the potential influence of the financial crisis.[4]

Mechanism analysis

To investigate how urban population size specifically impacts firm survival rates, we incorporate interaction terms between population size and key factors: openness, productivity cut-off, innovation level, and average wage. These factors provide a comprehensive understanding of how city size impacts firm survival through both agglomeration benefits and congestion costs, as shown in Table 2.[5]

Level of openness: Major cities in China typically engage in more foreign trade activities and attract a greater number of large foreign firms for investment. The coefficient of the interaction term is significantly positive, indicating that the vibrant foreign trade activities in large cities provide firms with more opportunities to secure international orders, thereby enhancing their survival rates.

Productivity cut-off: We calculate the lowest labour productivity in each city as the minimum productivity cut-off for firms operating there. Major cities in China generally have lower productivity cut-offs. Consequently, the negative significant interaction between city size and the productivity cut-off suggests that, compared to smaller cities, larger cities have more small entrants and lower productivity cut-offs.

Innovation: Larger cities tend to have higher levels of innovation. The empirical results reveal

[3]The detailed data and the specifications of the variables can be found in the Appendix A.
[4]The robustness of the results is confirmed, and detailed analyses are provided in Appendix B.
[5]We use a model to explain how four distinct mechanisms influence firm profit and survival theoretically. See Appendix C.

that the coefficient of the interaction between city size and innovation is significantly positive, indicating that the high levels of innovative activity in large cities enhance the probability of firm survival, consistent with the literature that identifies innovation as a key determinant of firm viability.

Average Wages: Average wages in major cities are typically higher, which increases the costs for firms. The coefficient of the interaction term is negative, suggesting that high wage levels decrease the probability of firm survival.

Taken together, the high levels of foreign trade activities, elevated innovation levels, and lower productivity cut-off in large cities contribute to an increased probability of firm survival, while high wage levels detract from it. This suggests that both agglomeration benefits, as reflected in trade, innovation, and productivity, and congestion costs, as reflected in rising wages, simultaneously influence firm survival.

Exploring the optimal city size

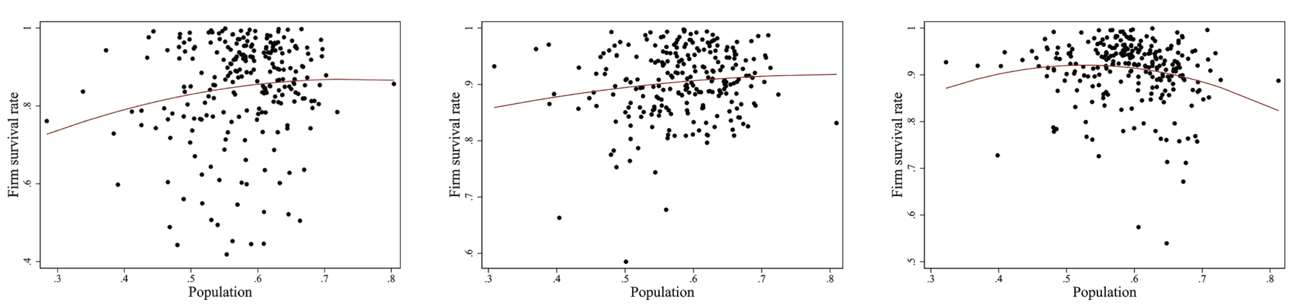
To better understand the interaction between these two opposing effects, we further explore the relationship between city size and firm survival. As shown in Figure 1, there is a possible 'inverted U' relationship between city size and firm survival rate. This would imply that both very small cities and megacities may not favour firm survival. We the add a quadratic term of city size to our regression model and the results are shown in Table 3.

Columns (1) and (2) of Table 3 show that the quadratic term is negatively significant, supporting the presence of an 'inverted U-shaped' relationship. At this optimal city population size, firm survival

Table 2. Mechanism analysis.

	(1)	(2)	(3)	(4)
Dependent variable	Firm survival rate			
City size	0.049**	0.038*	0.036*	0.140***
	(0.020)	(0.020)	(0.020)	(0.036)
City size × openness	0.001**			
	(0.001)			
City size × cutoff		−0.003*		
		(0.001)		
City size × innovation			0.0004**	
			(0.0002)	
City size × wage				−0.010***
				(0.003)
Observations	3150	2923	3150	3141
Controls	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Adj. R^2	0.110	0.104	0.107	0.112

Note: The brackets denote the standard error clustered at the city level; *$p < 0.1$, **$p < 0.05$, ***$p < 0.01$.



(a) 2001 (b) 2007 (c) 2014

Figure 1. Relationship between city size and survival rate of firms. Note: In Figure 1, we present a scatter plot of firm survival rate and city size for the years 2001, 2007, and 2014 to provide a preliminary analysis of their relationship. The maroon-coloured curve represents the fitted line, which indicates an inverted U-shaped relationship between firm survival rate and city size. The X-axis represents population size (in tens of millions), while the Y-axis denotes the firm survival rate. To mitigate the impact of outliers, we applied a 1% winsorization.

Table 3. Exploring the optimal city size.

	(1)	(2)	(3)	(4)
Dependent variable	Firm survival rate			
City size	0.082***	0.291*	0.077	0.349***
	(0.020)	(0.154)	(0.081)	(0.122)
City size^2	−0.007***	−0.023*	−0.003	−0.014***
	(0.002)	(0.013)	(0.004)	(0.005)
Observations	3471	3157	3237	2933
Controls	No	Yes	No	Yes
City FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Adj. R^2	0.008	0.085	0.035	0.105

Note: The brackets denote the standard error clustered at the city level; $^*p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

rates are maximized. By calculating the vertex of the quadratic term, we find that the optimal city population size for firm survival is around 5.59,[6] million people. In columns (3) and (4), we substitute total night-time lights for population size, and we find that the results remain robust.

IV. Conclusion

This paper explores the existence of an optimal city size that strikes a balance between the agglomeration benefits and congestion costs, aiming to maximize firm survival. Through empirical analysis using *Chinese Enterprise Registration Data* and city-level statistics, we find that firms in larger cities generally have higher survival rates. This is attributed to the positive externalities generated by large cities, such as increased openness, lower entry barriers, and enhanced innovation capacity, all of which help mitigate the adverse effects of higher wages and operational costs. However, the positive impact of city size on firm survival is not linear. Our results reveal an inverted U-shaped relationship between city size and firm survival from 2001 to 2014, with an optimal city size of 5.59 million people. Beyond this threshold, congestion costs surpass the agglomeration benefits, suggesting that excessively large cities may pose challenges for firm survival. These findings have significant implications for urban planning and policies aimed at promoting balanced urbanization and fostering sustainable business environments across city sizes.

Based on the study's conclusions, we propose the following policy implications: First, promote balanced urbanization by investing in infrastructure and innovation in medium-sized cities to alleviate pressure on megacities. Second, encourage entrepreneurship in medium-sized cities by optimizing financing channels and strengthening industry support to leverage their higher firm survival rates, creating a more favourable entrepreneurial environment.

Acknowledgements

We would like to thank Prof. Qiangyuan Chen for his valuable comments during the early stage of this project.

Author contributions

CRediT: **Haoyun Zhao:** Conceptualization, Data curation, Methodology, Project administration, Software, Supervision, Visualization, Writing – original draft, Writing – review & editing; **Zeyi Qian:** Data curation, Funding acquisition, Methodology, Project administration, Resources, Software, Visualization, Writing – original draft, Writing – review & editing; **Yameng Guo:** Supervision; **Yang Ye:** Visualization.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by the funding [202106890025] sponsored by the China Scholarship Council.

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[6]Based on the results in column (2) of Table 3, the optimal city size is calculated as $e^{0.291/(0.023\times2)} = 558.97$(measured in ten-thousands).

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